

Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)



Domain and range of arcsin

- 1 State the domain and range of $y = \arcsin x$.
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Domain and range of arccos

- 2 State the domain and range of $y = \arccos x$.
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Domain and range of arctan

- 3 State the domain and range of $y = \arctan x$.
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Evaluating inverse trigonometric functions exactly

- 4 Find the exact value of $\arcsin(\sqrt{3}/2)$, in radians.
- 5 Find the exact value of $\arccos(-1/2)$, in radians.
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Differentiating arcsin using the chain rule

- 6 Differentiate $y = \arcsin(3x)$ with respect to x .
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Differentiating arctan using the chain rule

- 7 Differentiate $y = \arctan(x^2)$ with respect to x .
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Differentiating a product involving arccos

- 8 Differentiate $y = x\arccos(x)$ with respect to x .
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Finding a gradient at a point

- 9 Find the gradient of the curve $y = \arcsin(x/2)$ at $x = 1$.
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Deriving the derivative of arctan from first principles

- 10 If $y = \arctan x$, then $x = \tan y$. Differentiate implicitly with respect to x to show that $\frac{dy}{dx} = \frac{1}{1+x^2}$.
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Identifying the graph of arctan

- 11 Which of the following best describes the graph of $y = \arctan x$?
- a. An odd function with horizontal asymptotes $y = \pm \frac{\pi}{2}$
 - b. An even function with a vertical asymptote at $x = 0$
 - c. A periodic function with period 2π
 - d. A function with domain restricted to $[-1, 1]$
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Tangent line to an inverse trigonometric curve

- 12 Find the equation of the tangent line to $y = \arcsin x$ at $x = 0$.
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Evaluating arctan exactly

- 13 Find the exact value of $\arctan(1)$, in radians.
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Differentiating arctan of a linear expression

- 14 Differentiate $y = \arctan(5x)$ with respect to x .
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Differentiating arcsin of a cubic expression

- 15 Differentiate $y = \arcsin(x^3)$ with respect to x .
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Evaluating a derivative at a specific point

16 Find the gradient of $y = \arccos(2x)$ at $x = 0$.

Tangent line at a non-zero point

17 Find the equation of the tangent line to $y = \arctan x$ at $x = 1$.

Differentiating arctan of a reciprocal expression

18 Differentiate $y = \arctan(1/x)$ with respect to x , and evaluate the derivative at $x = 1$.

Solving an equation involving arcsin

19 Solve $\arcsin(x) = \pi/6$ for x .

Solving an equation involving arccos

20 Solve $\arccos(x) = 2\pi/3$ for x .

Explaining why an inverse trig expression is undefined

21 Explain why $\arcsin(2)$ is undefined.

Proving the arcsin-arccos complementary identity

22 Prove that $\arcsin(x) + \arccos(x) = \pi/2$ for every x in $[-1, 1]$.

Differentiating a product involving arctan

23 Differentiate $y = x^2 \arctan(x)$ with respect to x .

Evaluating arccos of a surd value

- 24 Find the exact value of $\arccos(\sqrt{3}/2)$, in radians.
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A composite that does not simplify to the inner angle

- 25 Find the exact value of $\arcsin(\sin(2\pi))$, and explain why it is not equal to 2π .